Cloning of Human TFEB and Analysis of Its Nutrient-Dependent Subcellular Localization in HeLa Cells

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Version: Working draft

Date: October 23, 2026

Abstract

Transcription factor EB (TFEB) is a nutrient-responsive regulator of autophagy and lysosomal biogenesis. This proposal connects a defined human TFEB cloning design to a HeLa-cell localization assay. The selected reporter is TFEB–EGFP, with EGFP at the C terminus of TFEB. Reconstruction from versioned source sequences yields a 6,134 bp design encoding 722 residues, with an unmodified theoretical mass of 80.42 kDa. The proposed workflow uses high-fidelity PCR, HindIII/BamHI cloning into pEGFP-N1, bacterial screening, and complete fusion-region Sanger verification before cell use. The hypothesis is that starvation or Torin1 treatment increases reporter nuclear accumulation relative to the corresponding nutrient or solvent control. Fluorescence microscopy and anti-GFP immunoblotting will assess localization and expression. Sequence calculations are reported as in-silico checks; all experimental outcomes are predictions. The independent cell-assay details remain a part-A working draft.

Keywords: TFEB; autophagy; lysosome; mTORC1; EGFP fusion protein; subcellular localization

1 Rationale and Research Basis

TFEB is a member of the MiT/TFE transcription-factor family and a major transcriptional regulator of the autophagy–lysosome pathway. Under nutrient-rich conditions, mTORC1-dependent phosphorylation favors cytoplasmic retention of TFEB. During nutrient starvation or mTOR inhibition, TFEB accumulates in the nucleus and activates genes involved in lysosomal biogenesis and cellular clearance [26, 27]. Nutrient-dependent control of nuclear export also contributes to TFEB distribution [8].

HeLa cells expressing TFEB–GFP have already been used to measure nuclear accumulation after starvation or Torin1 treatment [26]. The present course project applies this established biological response to evaluate a newly designed reporter; it does not claim discovery of the mechanism. Its primary readout is the nuclear-to-cytoplasmic distribution of the fusion protein. Molecular identity must be established before that distribution can be interpreted.

1.1 Research Objectives and Hypothesis

The first objective is to construct and sequence-verify an EGFP-tagged TFEB mammalian expression plasmid. The second is to compare its subcellular localization under normal culture, starvation, and mTOR-inhibitor conditions. The hypothesis is that starvation increases the nuclear-to-cytoplasmic fluorescence ratio relative to time-matched nutrient-replete culture, and that Torin1 increases the ratio relative to its matched vehicle control.

2 Experimental Design

2.1 Overall Workflow

The study proceeds from reference-sequence retrieval and primer design to PCR amplification, plasmid construction, clone verification, transient transfection, and localization analysis. Sequence verification is the gate between molecular cloning and cell-based experiments.

2.2 Sequence-Defined Construct and Materials

The proposed construct, pEGFP-N1-TFEB-Bv1, will express human TFEB followed by a C-terminal EGFP tag under the CMV promoter. The reference will be TFEB transcript variant 1, NM_007162.3, whose CDS at transcript positions 303–1733 encodes NP_009093.1 (476 amino acids). A native, intron-free 1,431-nt CDS containing its original `TGA` will be ordered as linear double-stranded DNA without codon optimization, for example through the gBlocks HiFi synthesis service. This defined template will avoid uncertainty about the TFEB isoform in a cellular cDNA preparation. The synthesis feasibility, supplier QC and physical material identity will be checked before use; physical template and vector stocks have not been experimentally verified in this project [7, 9].

The empty backbone will be pEGFP-N1, GenBank U55762.1 (4,733 bp). Its bacterial kanamycin resistance and mammalian neomycin/G418 selection have different uses. The reference map includes the CMV promoter, EGFP, polyadenylation signals and replication origins, but the manufacturer's reference sequence does not certify a physical stock [1, 6]. C-terminal EGFP has a precedent in TFEB localization experiments; the present HindIII/BamHI design is not claimed to be sequence-identical to Addgene #38119 or its historical HindIII/KpnI construction [5, 26].

The accepted *in silico* design is 6,134 bp (Figure 1). It retains the native TFEB start codon, removes the TFEB stop through reverse-primer placement, and connects TFEB to EGFP through the seven-residue `GDPPVAT` linker. EGFP's `ATG` remains an internal methionine codon, and its terminal `TAA` ends the 722-residue fusion. The forward primer adds `ACC` immediately before TFEB's start, giving an A at position -3 and G at $+4$. Reverse-primer `CC` contributes coding-strand `GG` to preserve the reading frame. Removing a stop alone would not establish this frame; both junctions and the entire fusion were checked computationally.

The proposed laboratory workflow (Figure 2) will use Q5 High-Fidelity $2\times$ Master Mix (NEB M0492) for cloning PCR, HindIII-HF R3104 and BamHI-HF R3136 in rCutSmart, Monarch cleanup and gel-extraction kits, T4 DNA Ligase M0202, NEB 5-alpha C2987 competent *E. coli* (a DH5α derivative), Monarch T1110 minipreps and Sanger primer walking. OneTaq M0482 will be used only for screening; its products will not enter the construct.

2.3 Cloning Primers and High-Fidelity PCR

Table 1 gives the complete oligonucleotides and their functional parts. The forward annealing segment matches CDS positions 1–22; the reverse segment is the reverse complement of CDS positions 1407–1428, excluding the native stop. HindIII and BamHI have no internal sites in the selected TFEB coding segment and provide distinct $5'$ cohesive ends. Six protective bases per primer are a conservative choice: the manufacturer's end-cleavage table assigns both enzymes its highest category at three flanking base pairs, but that category represents 50–100% cleavage under the stated oligonucleotide assay conditions, not guaranteed complete digestion of this amplicon [10, 11, 15].

Table 1: Complete cloning primers and sequence-derived properties. Every sequence is written 5′ to 3′. In the partitioned lines, the order is protective bases, recognition site, deliberate addition, and template-annealing segment.

Primer	Sequence and properties
TFEB-HindIII-F	`ACTGACAAGCTTACCATGGCGTCACGCATAGGGTTGC`
	Parts: `ACTGAC` | `AAGCTT` | `ACC` | `ATGGCGTCACGCATAGGGTTGC`
	Full: 37 nt, 54.054% GC; annealing segment: 22 nt, 59.091% GC, $T_m = 69.498\,^{\circ}\mathrm{C}$. Full-oligo $T_m = 77.238\,^{\circ}\mathrm{C}$, not used for initial annealing.
TFEB-BamHI-R	`TATGACGGATCCCCCAGCACATCGCCCTCCTCCATG`
	Parts: `TATGAC` | `GGATCC` | `CC` | `CAGCACATCGCCCTCCTCCATG`
	Full: 36 nt, 61.111% GC; annealing segment: 22 nt, 63.636% GC, $T_m = 69.050\,^{\circ}\mathrm{C}$. Full-oligo $T_m = 78.896\,^{\circ}\mathrm{C}$, not used for initial annealing.

The reported estimates use Primer3-py 2.4.0 with SantaLucia thermodynamics and salt correction, 50 mM monovalent ions, 2.0 mM Mg^{2+}, 0.8 mM total dNTP and 500 nM primer. The monovalent-ion concentration is a modeling assumption, not a claim about Q5's proprietary buffer. The complete annealing segments and their terminal 12-mers each have one exact site in the defined template. Full-oligo hairpin and dimer screening passed the prespecified $55\,^{\circ}\mathrm{C}$ cutoff. These checks do not establish transcriptome-wide specificity or experimental amplification performance [25]. Only template-complementary segments will guide initial PCR annealing, because the added tails do not bind the original template.

Reactions will be assembled on ice using Table 2. The 1 ng template input is a design target for the synthetic linear template; the working dilution will be calculated from measured concentration. A no-template control will replace the template with water [24].

Table 2: Proposed Q5 cloning PCR, 50.0 μL total. Stock targets are not measured concentrations.

Component	Volume	Final concentration or input
Q5 M0492 $2\times$ Master Mix	25.0 μL	$1\times$; Mg^{2+} 2.0 mM; each dNTP 200 μM
TFEB-HindIII-F, 10 μM	2.5 μL	0.5 μM
TFEB-BamHI-R, 10 μM	2.5 μL	0.5 μM
Synthetic template, target 1 ng/μL	1.0 μL	1 ng total
Nuclease-free water	19.0 μL	To 50.0 μL

The initial program will be $98\,^{\circ}\mathrm{C}$ for 30 s; 30 cycles of $98\,^{\circ}\mathrm{C}$ for 10 s, $65\,^{\circ}\mathrm{C}$ for 20 s and $72\,^{\circ}\mathrm{C}$ for 45 s; then $72\,^{\circ}\mathrm{C}$ for 2 min. The 45 s extension rounds the expected 1,457-bp product upward at 30 s/kb. Initial annealing at $65\,^{\circ}\mathrm{C}$ and a $62/65/68\,^{\circ}\mathrm{C}$ optimization gradient are design choices, not Q5-calculator outputs or measured optima [24].

2.4 Product Assessment, Restriction Digestion and Ligation

A 5 μL PCR aliquot will be analyzed on 1.0% agarose in $1\times$ TAE with $1\times$ loading dye, a compatible DNA stain and 0.5 μg NEB N3232 1 kb ladder. An initial field of 5 V/cm of electrode separation is a design choice; electrophoresis will stop after the relevant bands separate and before the tracking front

exits. The ladder's 500/517 bp pair is expected to co-migrate. Uncut plasmids will not be sized by comparison with linear DNA, and illumination of DNA intended for cloning will be minimized [20].

A single 1,457-bp PCR product will be purified with Monarch Spin PCR & DNA Cleanup T1130: 20–100 μL sample, five sample volumes of prepared BZ, an S1A column, two 200 μL WZ washes, and 20 μL EY elution after a 1 min wait. Centrifugation will follow the current kit protocol at $16{,}000 \times g$ and room temperature. If additional bands occur, the target band will be excised and purified with T1120 using four gel-slice volumes of BY, dissolution near $50\,^{\circ}$C, an S1A column, two 200 μL WZ washes and 20 μL EY elution. Alcohol-containing buffers will be prepared according to their current bottle labels. Recovered DNA will be measured; no yield or recovery fraction is assumed [16, 17].

PCR product and empty vector will be digested separately with the common recipe in Table 3, at $37\,^{\circ}$C for 60 min. Both enzymes have full listed activity in rCutSmart; the proposed enzyme volumes contribute 2% glycerol. Actual DNA and water volumes will be recalculated from measured concentrations. Insufficient material will prompt repeat or scaled preparation. Uncut and single-digest analytical controls will be included when troubleshooting incomplete digestion. BamHI-HF cannot be heat-inactivated, so digestion products will be purified before ligation [10, 13, 15].

For direct preparative or diagnostic gel loading of unpurified BamHI-HF digests, SDS-containing Purple Gel Loading Dye B7024 ($6\times$) will be diluted to $1\times$ by adding one volume of dye to five volumes of sample. This will dissociate residual enzyme-bound DNA; the no-SDS B7025 product will not substitute for it. Dye will be added only to gel samples. If SYBR Safe or GelRed is used, staining will occur after electrophoresis as directed for B7024 [10, 14].

The 4,695-bp backbone will be gel-purified to separate residual circular vector and the excised 38-nt MCS. The digested insert will be purified with T1130. Its nominal retained length is 1,439 nt; the discarded top-strand terminal pieces are 7 and 11 nt. These short pieces and the 38-nt MCS will not be expected to resolve usefully on the proposed gel. Both retained fragments have equal strand lengths; their pydna duplex spans are 4,699 and 1,443 bases because of staggered ends. Gel sizes will use the nominal top-strand cut-to-cut lengths, and molarity will use mean strand lengths, which equal those nominal lengths for the selected ends. Distinct $5'$ `AGCT` and `GATC` overhangs will support directional ligation. Routine dephosphorylation will be omitted; vector-only background will be used to assess incomplete digestion or circular-vector carryover.

The initial ligation will contain 20 fmol backbone and 60 fmol insert (3:1 insert:vector). With mean strand length L, the mass conversion is

$$m\ (\mathrm{ng}) = n\ (\mathrm{fmol}) \times L\ (\mathrm{nt}) \times 0.00066.$$

The calculated inputs are 61.974 ng backbone and 56.9844 ng insert. Table 3 gives a practical recipe rounded to 0.01 μL. The unrounded total DNA concentration is 5.948 ng/μL, within the manufacturer's 1–10 ng/μL starting range. Fully thawed ATP-containing ligase buffer will be mixed before use. Ligation will proceed at 20–25 °C for 10 min, followed by $65\,^{\circ}$C for 10 min. This last treatment terminates the ligation reaction; it does not replace purification of the preceding BamHI-HF digests. A matched vector-only ligation will replace the insert volume with water. No polynucleotide kinase is needed after the planned restriction digestion [12].

Table 3: Proposed digestion and ligation mixtures. DNA stock concentrations are preparation targets to be adjusted using measurements.

Component	Volume	Amount or final condition
Each separate digestion: 50.0 μL		
DNA, target 50 ng/μL	10.0 μL	500 ng
$10\times$ rCutSmart	5.0 μL	$1\times$
HindIII-HF, 20 U/μL	1.0 μL	20 U
BamHI-HF, 20 U/μL	1.0 μL	20 U
Water	33.0 μL	To 50.0 μL
Ligation: 20.00 μL		
Backbone, target 20 ng/μL	3.10 μL	Approximately 20 fmol
Insert, target 10 ng/μL	5.70 μL	Approximately 60 fmol
$10\times$ T4 DNA Ligase Buffer	2.00 μL	$1\times$
T4 DNA Ligase M0202, 400 U/μL	1.00 μL	400 U
Water	8.20 μL	To 20.00 μL

2.5 Transformation, Screening and Plasmid Preparation

For each transformation, 2 μL ligation will be added to 50 μL NEB 5-alpha C2987 cells. Cells will remain on ice for 30 min, undergo heat shock at $42\,^{\circ}$C for 30 s, return to ice for 5 min, and receive 950 μL SOC. Recovery will proceed at $37\,^{\circ}$C and 250 rpm for 60 min. Then 100 μL each of undiluted recovery and 1:10 and 1:100 SOC dilutions will be plated on LB-kanamycin and incubated at $37\,^{\circ}$C overnight. Kanamycin will be 30 μg/mL in bacterial plates and liquid cultures; G418 will not be used for bacterial cloning [21, 22].

Vector-only ligation, a no-DNA control and the supplied pUC19 transformation control will be processed in parallel. pUC19 will be selected on ampicillin at 100 μg/mL. These controls will assess backbone background, contamination or selection failure, and competent-cell handling, respectively. The pUC19 control will not test kanamycin resistance or the ligation itself. No transformation efficiency or fraction of correct clones is assumed [21].

Eight isolated colonies will initially be screened while numbered patches are retained. Each 25 μL OneTaq M0482 reaction will receive trace colony material, 12.5 μL $2\times$ mix, 0.5 μL of each 10 μM diagnostic primer and 11.5 μL water; final primer concentrations will be 0.2 μM. Colony lysis will be $94\,^{\circ}$C for 5 min, followed by 30 cycles of $94\,^{\circ}$C for 30 s, $58\,^{\circ}$C for 30 s and $68\,^{\circ}$C for 160 s, then $68\,^{\circ}$C for 5 min. The 160 s extension covers the longest modeled 2,608-bp product at at least 1 min/kb. An initial $58\,^{\circ}$C annealing step and $55/58/61\,^{\circ}$C gradient are design choices [23].

S-F1/S-R1 will flank the insert, while S-F2/S-R1 will combine an internal TFEB primer with a downstream vector primer to test orientation (sequences and sites in Table 4). Their model T_m values are $60.04/62.55\,^{\circ}$C and $66.09/62.55\,^{\circ}$C, respectively, using 50 mM monovalent ions, 1.8 mM Mg^{2+}, 0.8 mM total dNTP and 200 nM primer. This is separate from the Sanger-primer model below [25]. The correct-construct predictions are 2,605 bp for flanking PCR and 2,093 bp for orientation PCR; an empty vector predicts 1,204 bp for flanking PCR and no specific orientation product. Empty-vector and no-template controls will accompany screening. A missing orientation band will remain provisional without a functioning positive control: in the first construction round, flanking-positive minipreps will

proceed to sequencing if all orientation assays are negative. A sequence-accepted clone may provide the positive control in later rounds.

Up to three screening-positive colonies will be grown independently in 3 mL LB-kanamycin at 37 °C for 12–16 h with aeration; if fewer are recovered, those available will be analyzed. For each T1110 miniprep, 1.5 mL culture will be processed with 200 μL RNase-containing B1, 200 μL B2 for 1 min with gentle inversion, then 400 μL B3 for 2 min. After clarification, DNA will bind to an S2D column, receive 200 μL BZ and 400 μL WZ washes, and elute in at least 30 μL EY. For DNA intended for transfection, the manufacturer's additional 5 min BZ incubation will be used; this is not an endotoxin-free certification. Yield and purity will be measured [18, 19].

A 500-ng miniprep aliquot will undergo the defined 50-μL HindIII/BamHI digestion and gel analysis with the B7024 handling above. Gross agreement with 4,695- and 1,439-bp fragments will support the insert/backbone sizes, but will not exclude reversed insert models, short indels or substitutions. A three-base retained-stop difference will not be resolved reliably on the proposed gel. Complete target sequencing will be the acceptance gate.

2.6 Sanger Primer Walking and Acceptance Criteria

Each candidate will be sequenced with the ten primers in Table 4, in separate reactions. For Eurofins Standard Read purified-plasmid submissions, this construct falls in the 150–250 ng/μL stock range. A target of 200 ng/μL and 55 μL DNA per clone will support ten reads in the separate-template format ($10 + 9 \times 5$ μL). If needed, independently prepared aliquots from the same single-colony culture will be pooled and measured or concentrated; different clones will not be pooled. Each primer will be supplied separately at 10 μM, with at least 10 μL for its first read and 5 μL per additional use. These are submission targets, not claimed miniprep yields; current service requirements will be rechecked before ordering [2].

Sanger-primer estimates use Primer3/SantaLucia with 50 mM monovalent ions, no added Mg^{2+} or dNTP, and 50 nM oligo. This reproducible comparison is not the sequencing service's proprietary chemistry. Selection required 18–25 nt, 35–60% GC, T_m of 50–62 °C, no four-base homopolymer, a terminal G/C without three consecutive terminal G/C bases, and modeled hairpin/homodimer T_m < 40 °C. Each oligo has one exact full-length binding site in the final construct; genome-wide specificity has not been tested [2, 25].

Table 4: Sanger and diagnostic primers. Coordinates are 1-based inclusive on pEGFP-N1-TFEB-Bv1. A plus/minus sign denotes the extension direction; every oligonucleotide is written 5′ to 3′. Windows are predictions, not measured reads.

ID	Complete sequence	nt; GC%; T_m (°C)	Binding site (strand)	Planned usable window
S-F1	GGGATTTCCAAGTCTCCACC	20; 55.00; 52.03	418–437 (+)	488–1087
S-F2	TGGGAACAAGTTTGCTGCCCAC	22; 54.55; 58.13	930–951 (+)	1002–1601
S-F3	ACCATCCTCAAGGCCTCTGTGG	22; 59.09; 57.96	1456–1477 (+)	1528–2127
S-F4	TGATCCACTTCTGTCCACCATGTCC	25; 52.00; 58.23	1971–1995 (+)	2046–2645
S-F5	GCTGGAGTACAACTACAACAGCCAC	25; 52.00; 57.83	2502–2526 (+)	2577–3176
S-R1	GACAAACCACAACTAGAATGCAGTG	25; 44.00; 54.94	2998–3022 (−)	2348–2947
S-R2	GTTGCCGTCCTCCTTGAAGTCG	22; 59.09; 57.87	2466–2487 (−)	1816–2415
S-R3	ATGGTGGACAGAAGTGGATCAGAGG	25; 52.00; 57.76	1967–1991 (−)	1317–1916
S-R4	ATCCACAGAGGCCTTGAGGATGG	23; 56.52; 57.86	1457–1479 (−)	807–1406
S-R5	TGATGTGGGCAGCAAACTTGTTCC	24; 50.00; 58.35	932–955 (−)	282–881

The conservative planning window comprises extended bases 51–650 after each primer's 3′ end (600 usable bases), excluding the early low-quality region and remaining within a standard-read length. Adjacent planned windows overlap by at least 50 nt. The acceptance target is positions 602–2799 (2,198 nt): 20 upstream vector bases, the 5′ junction and initiation context, and the full 2,169-nt fusion ORF through its final stop. The computational coverage check gives every target base at least one forward and one reverse read; primer-binding bases within this target are covered by other primers. The outer S-F1 and S-R1 sites lie outside the acceptance target. This plan does not establish bidirectional sequencing of those external sites or of the complete vector backbone [2, 4].

A clone will be accepted only after actual quality-trimmed chromatograms provide unambiguous, concordant Q20-or-better calls in both directions across the entire target, no unexpected nucleotide changes, and the predicted restriction pattern. Reads will be aligned to the exact B-v1 reference, with manual inspection of junctions and all mismatches. Q20 is a per-base error-probability metric, not a guarantee that a molecule is error-free [3]. Low-quality regions, single-strand gaps and ambiguous peaks will trigger additional reads or primers. Reproducible mixed peaks will prompt single-colony re-isolation and a fresh preparation. A clear unwanted coding or junction mutation will reject that clone; a shared error across independent clones will prompt investigation of template and design. Predicted coverage will never substitute for observed trace quality.

2.7 Troubleshooting and Experimental Boundary

No cloning-PCR band will prompt checks of template identity, dilution, primer matching and annealing conditions; additional bands will prompt the planned gradient and target-band purification. High vector-only background will prompt checks for incomplete digestion and residual circular DNA. Absence of colonies in both the test and positive transformation control will point to handling or selection problems before failed ligation is inferred. An apparently correct gel pattern with sequence disagreement will fail acceptance. Only sequence-accepted plasmid will enter the proposed HeLa experiment; expression, localization and biological behavior will still require cellular validation.

2.8 Cell Culture, Transfection, and Treatment

After experimental sequence acceptance, pEGFP-N1-TFEB-Bv1 (TFEB–EGFP) will be transiently transfected into HeLa cells. Starvation will be compared with time-matched nutrient-replete culture. Torin1-treated cells will be compared with nutrient-replete cells receiving the same final solvent concentration. The corresponding empty pEGFP-N1 (U55762.1; 4,733 bp) will serve as the EGFP-only vector control; mock-transfected cells will define background. These controls will be included under the corresponding treatments. The final cell-assay protocol will identify the culture medium, transfection reagent, cell density, DNA amount, and treatment concentration and duration. These independent part-A parameters remain to be finalized; no cell experiment is claimed here.

2.9 Imaging and Localization Analysis

After treatment, cells will be fixed and counterstained with DAPI. Fluorescence images will be collected using consistent acquisition settings. Nuclear and cytoplasmic regions will be defined for each eligible transfected cell, and background-corrected mean fluorescence will be used to calculate the nuclear-to-cytoplasmic ratio:

$$R_{\mathrm{N/C}} = \frac{I_{\mathrm{nucleus}} - I_{\mathrm{background}}}{I_{\mathrm{cytoplasm}} - I_{\mathrm{background}}}. \tag{1}$$

At least three independent experiments are planned. The number of cells per replicate, eligibility criteria, and handling of background-level cytoplasmic signals will be specified before analysis. Cells within one experiment will not be treated as independent biological replicates. The final statistical comparison will account for the experimental unit and the planned control-versus-treatment comparisons.

2.10 Expression Validation

An anti-GFP Western blot will assess expression of the 722-residue fusion and additional immunoreactive species. Its calculated unmodified average mass is 80.42 kDa, including the GDPPVAT linker. This value guides interpretation and is not an exact SDS-PAGE migration criterion: TFEB phosphorylation-dependent mobility shifts have been reported [26]. The EGFP-only vector expresses the distinct 239-residue tag product. A lower band in the fusion sample would prompt checks for cleavage or other products rather than an identity assignment from size alone. The final part-A protocol must specify lysis, loading amount/control, antibodies and detection conditions. Whole-cell immunoblotting assesses expression; the imaging assay establishes localization.

3 Expected Results and Criteria for Success

3.1 Computed Construct and Experimental Predictions

The in-silico pEGFP-N1-TFEB-Bv1 (6,134 bp; Figure 1) encodes a 722-residue TFEB–EGFP fusion: 476 TFEB residues, the seven-residue GDPPVAT junction, and 239 EGFP residues. The native TFEB stop is absent; EGFP supplies the final TAA. The calculated unmodified average mass is 80.42 kDa.

The reconstruction and an independent standard-library audit agreed on sequence, reading frame, junctions, terminal stop and fragment geometry. Corresponding checks rejected four software challenges: reversed TFEB, a one-base frameshift, restored native TFEB stop, and an internal HindIII site. These are not experimental clones or a measured success rate.

Cloning PCR predicts a 1,457 bp product. HindIII/BamHI digestion yields a 1,439 nt insert and 4,695 nt backbone by the cut-to-cut single-strand convention. Their pydna duplex spans are 1,443 and 4,699 bp, including four-base overhangs; gel sizes use the nominal single-strand lengths. Because the empty vector loses a 38 nt MCS segment, the final length is not the intact vector length plus the undigested amplicon.

Table 5 and Figure 3 summarize screening. Flanking PCR distinguishes insert-bearing candidates from empty vector. Interpret a negative orientation assay only with an amplification-positive control. If all orientation assays are negative in the first round, sequence flanking-positive candidates. Reverse TFEB is a hypothetical challenge, not an expected product of incompatible-end directional ligation.

Flanking PCR and HindIII/BamHI digestion cannot exclude reversed TFEB; the proposed gel cannot resolve three-base retained-stop differences. Acceptance requires the expected digest and actual, unambiguous, concordant bidirectional sequence across 602–2799 (2,198 nt), including both junctions, the complete fusion ORF and terminal stop. Ten primers predict usable coverage in both directions at every target base, not experimental coverage or whole-backbone verification. Resolve low-quality calls, mixed peaks, gaps and unexpected target changes under the Sanger acceptance procedure before cell assays. Figure 2 summarizes the planned workflow and controls.

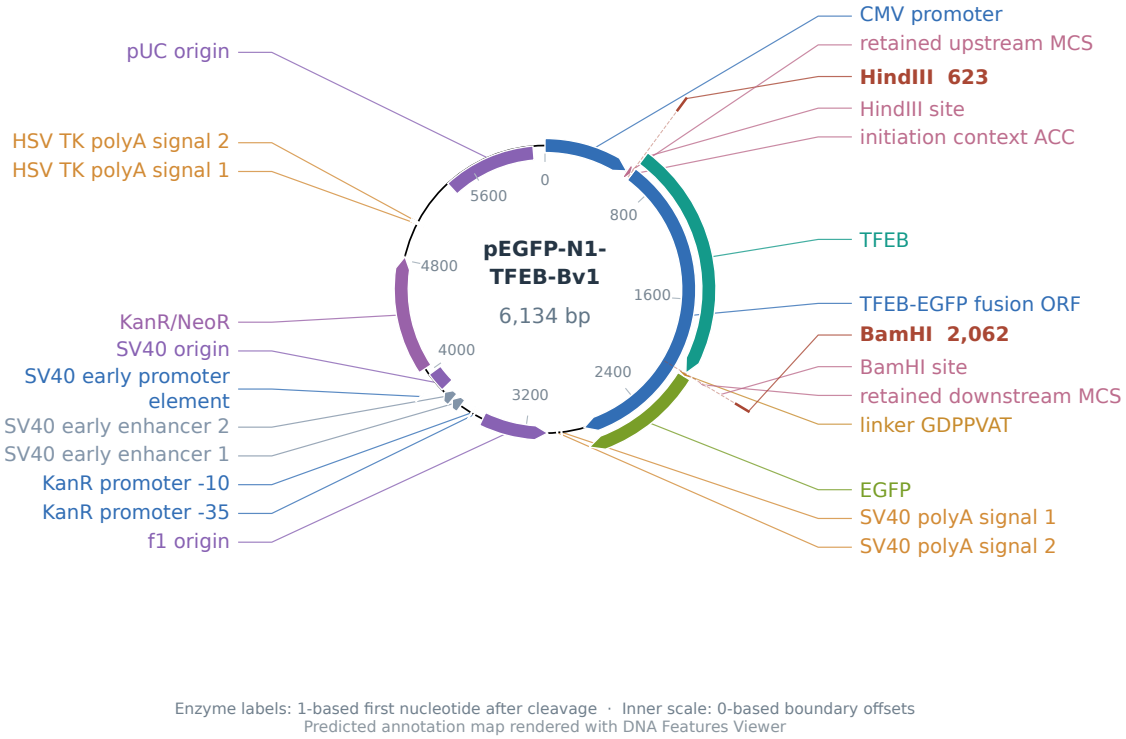


Figure 1: Predicted annotation map of pEGFP-N1-TFEB-Bv1 (6,134 bp), rendered from the accepted GenBank sequence. The complete fusion ORF overlaps its TFEB, linker and EGFP components intentionally. HindIII 623 and BamHI 2062 denote the first reference-strand nucleotide after cleavage; their recognition sequences begin at 622 and 2061, respectively. Poly(A) annotations are signal motifs, not complete processing regions. KanR applies to bacteria and Neo/G418 to mammalian selection. The pUC and SV40 origins are shown without an inferred directional arrow. No experimental observation is represented.

Table 5: Predicted screening products. All sizes are nominal bp; "none" means no specific product is predicted. R and S are hypothetical challenge sequences.

Sample	Flanking PCR (S-F1/S-R1)	Orientation PCR (S-F2/S-R1)	HindIII/BamHI digest
Intended construct (C)	2,605	2,093	4,695 + 1,439
Empty vector (E)	1,204	none	4,695 + 38
Reverse TFEB (R)	2,605	none	4,695 + 1,439
Retained TFEB stop (S)	2,608	2,096	4,695 + 1,442

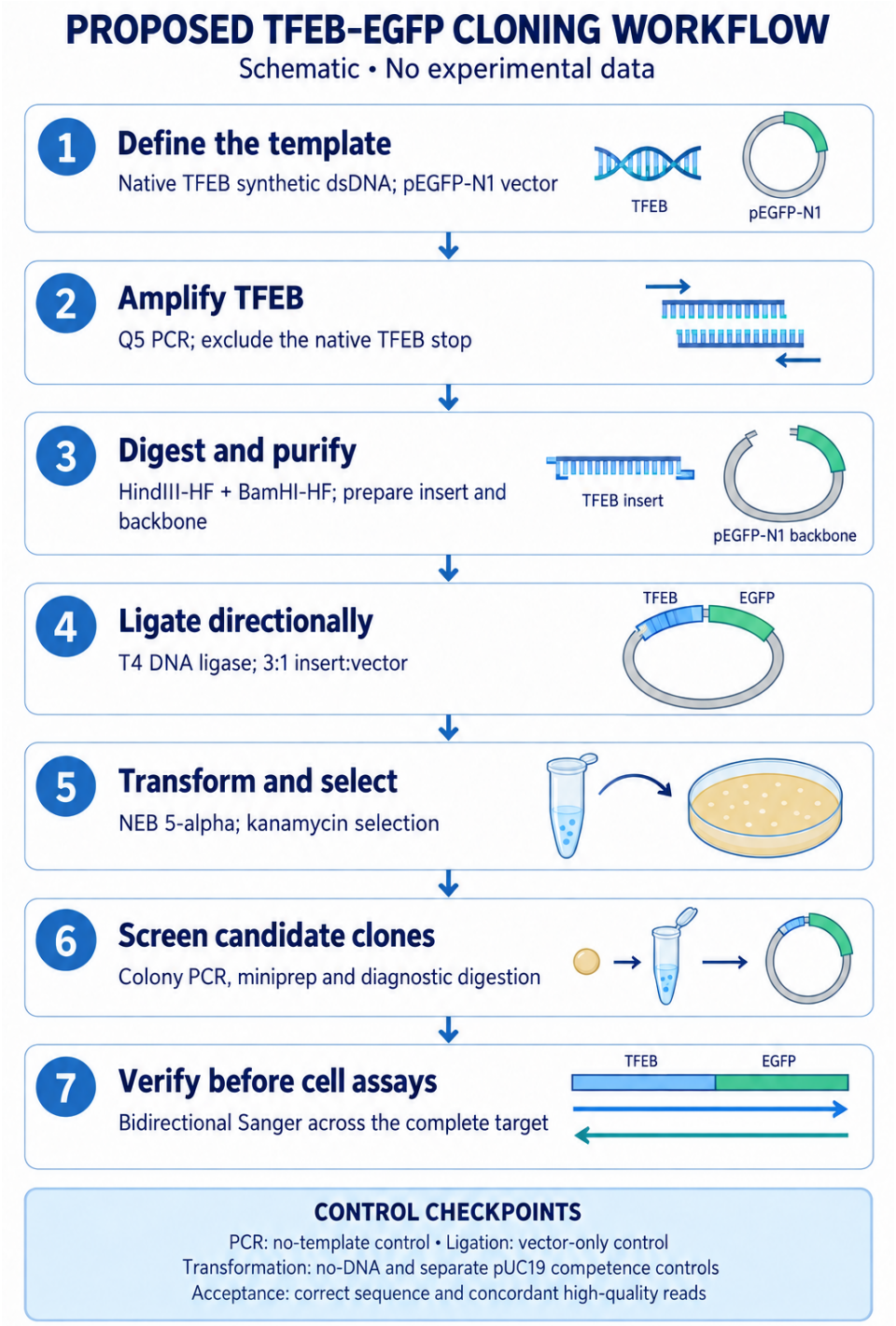


Figure 2: Proposed cloning workflow and control checkpoints. AI-assisted illustrations are symbolic, not exact molecular geometry, plasmid coordinates or experimental data. The full-length paired arrows summarize the intended bidirectional sequencing coverage, not observed reads. Each operation is planned; measured DNA concentrations and actual sequence quality are required before clone acceptance.

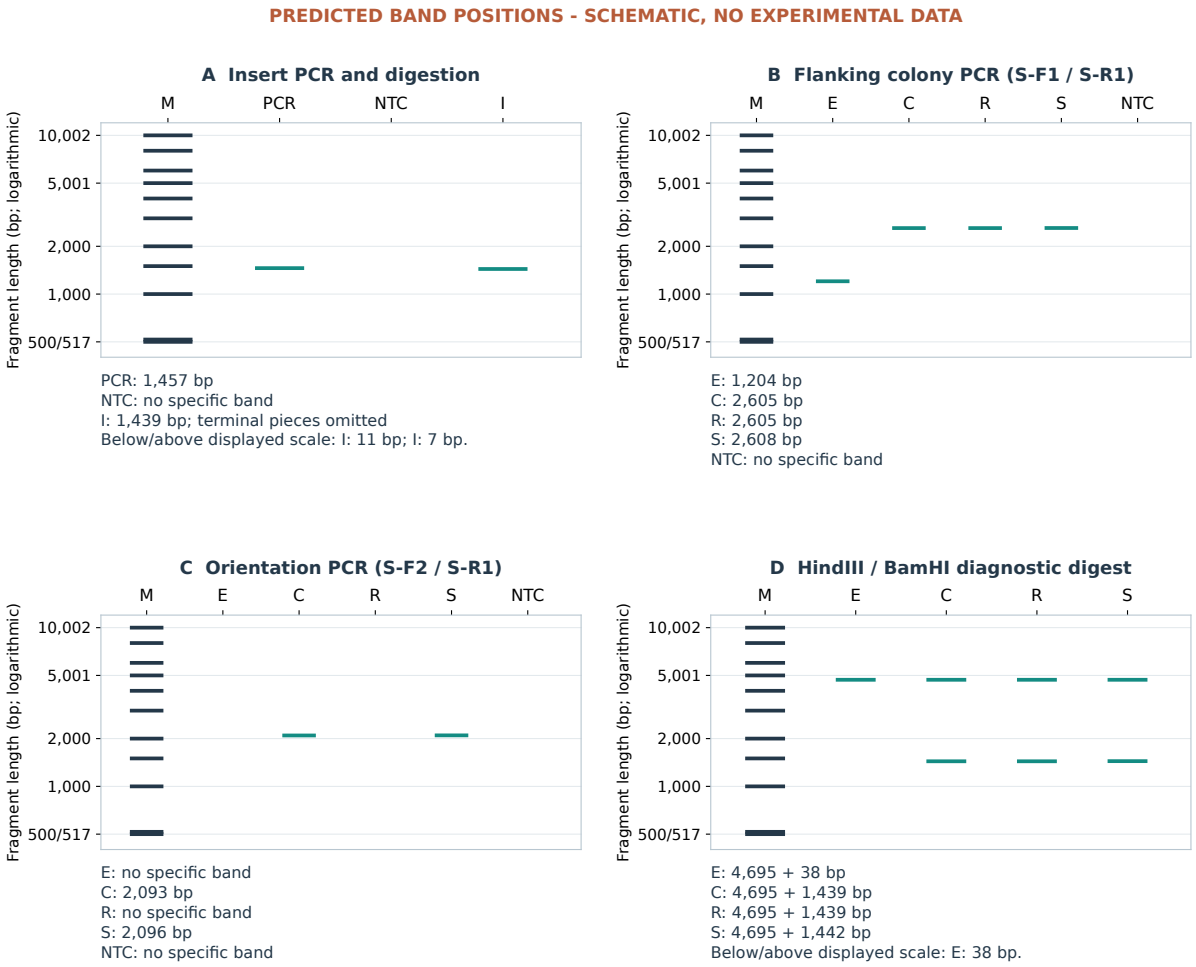


Figure 3: Predicted DNA-band positions, drawn as a schematic rather than an experimental gel. M: NEB N3232 ladder; PCR: tailed insert amplicon; I: digested insert; NTC: no-template control; E: empty vector; C: intended construct; R: hypothetical reverse-TFEB model; S: hypothetical retained-stop model. The 500/517 bp marker pair is not resolved [20]. The 7, 11 and 38 nt fragments fall below the displayed range. Three-base differences between C and S are not expected to resolve on the proposed 1% gel. An empty NTC schematic means no specific product is predicted, not a guarantee of a clean experimental lane. Uncut circular DNA mobility and band intensity are not predicted.

3.2 Expression and Nutrient-Dependent Localization

The TFEB–EGFP reporter is expected to show a higher nuclear-to-cytoplasmic fluorescence ratio after starvation than in time-matched nutrient-replete culture, and after Torin1 than with the matched solvent control. No effect size or statistical significance is assumed. A principal anti-GFP-reactive species consistent with an approximately 80.42 kDa fusion, interpreted with mobility and expression controls, would support production of the intended reporter.

Evidence for nutrient responsiveness will require a reproducible treatment effect across independent experiments, supported by expression and imaging controls. The magnitude and uncertainty of the localization change will be reported alongside the statistical comparison. A failure to observe redistribution would prompt checks of sequence, fusion orientation, expression level, treatment response, and imaging conditions.

4 Discussion and Limitations

The proposed design links molecular identity to a testable localization phenotype. A correct DNA sequence is necessary but does not establish that the fusion preserves endogenous TFEB regulation. Fluorescent tagging and overexpression may alter abundance, interactions, and distribution. The matched EGFP-only vector controls tag-associated fluorescence and background; it does not prove that the fusion is functionally equivalent to endogenous TFEB. An absence of redistribution would therefore require checks of construct identity, intact protein expression, treatment response, and acquisition conditions before rejecting the biological hypothesis.

The localization assay measures redistribution rather than autophagic flux or lysosomal degradative activity. Nuclear accumulation can also accompany lysosomal dysfunction [27]; it cannot alone establish increased clearance. Moreover, an endpoint nuclear-to-cytoplasmic ratio does not separate increased import from reduced export, because TFEB export is itself regulated [8]. Comparison with endogenous TFEB or additional functional assays would address these limitations, but they are outside this basic cloning-and-localization design.

Author Contributions

The agreed division of responsibility assigns molecular cloning and construct design (part B) to Siyuan Jiang, and the biological rationale and cell-assay design (part A) to Ziyan Song. This is a planned division, not a record of completed individual contributions. The present B work includes the A content needed to connect the construct to expression and localization assays, with Siyuan Jiang's authorization. Each author must confirm their actual contributions and review the integrated manuscript before submission. No wet-laboratory work is reported.

AI Use Declaration

AI assistance (OpenAI Codex) was used for literature checking, sequence-design code, local computational checks, drafting and integration, figure preparation, and LaTeX compilation and layout review. The workflow illustration was generated and revised with OpenAI ImageGen, then checked against the accepted design; prompts and source images are retained. The coordinate map and predicted band positions were generated by code from verified sequence and fragment tables. Computational predictions

and checks are identified separately from proposed experimental outcomes; AI-generated text or graphics do not represent experimental observations. Source records, calculation inputs, and reproducible scripts accompany the design. The authors remain responsible for verifying the scientific claims, practical feasibility, final parameters, and contribution statement before submission.

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